Airbnb Clone Guide

Soft Light | Warm Sepia | Dark Dimmed

SOFT LIGHT EDITION • ZERO GLARE

Project Architecture, Workflows & Engineering Solutions

A calm, comfortable visual guide designed for effortless reading. Explains how our Airbnb clone works, every engineering challenge faced, why it happened, and how we solved it with simple block diagrams and fitted web screenshots.

- Next.js 16.4 (App Router)
- React 19
- Tailwind CSS v4
- Framer Motion
- Strict TypeScript 5
- 100% Mobile Responsive
- Turbopack (< 2.5s builds)

QUICK NAVIGATION & INDEX

Section 1 System Architecture
Feature-sliced domain tree & responsibilities

Section 2 User Workflows
Header morphing, booking math & modal flows

Section 3 Visual Feature Map
8 fitted screenshots with routes & components

Section 4 Engineering Problems (1–8)
Header alignment, 404 recovery, back button & host sizing

Section 5 Build & Verification
Commands, TypeScript health & checklist

1 Big Picture Architecture (How Code is Organized)

We use **Feature-Sliced Domain Architecture**. Code is cleanly separated into modular features instead of being dumped into one large folder.



Feature Directory	What Lives Inside?	Why It Is Kept Separate
`src/features/home/`	HomeHeader, ListingCarousel, DestinationCarousel, InspirationSection	Keeps the discovery feed and category tabs isolated from room details.
`src/features/listing/`	PhotoBentoGrid, PropertyOverview, ListingHeader, SubNavbar	Encapsulates the 5-photo bento grid and room specifications.
`src/features/photo-tour/`	PhotoTourOverlay, 43 categorized photos across 9 room sections	Manages the full-screen photo modal with sticky floating headers.
`src/features/lightbox/`	LightboxModal, keyboard navigation hook (Esc, ←, →)	Focuses on single-photo viewing with zero clutter.
`src/features/booking/`	StickyBookingWidget, dynamic price calculation hook	Recalculates nights, base rate, fees, and grand total in real-time.

2 Complete End-to-End User Workflows

Here is what happens step-by-step when a user navigates and interacts with the app:

Workflow A: Home Page Scroll & Morphing

HOME PAGE

When you load `localhost:3000/`, the header starts tall (180px) with 3D icons. As you scroll down, it gracefully morphs into an 80px compact pill.

HEADER MORPHING FLOW DIAGRAM

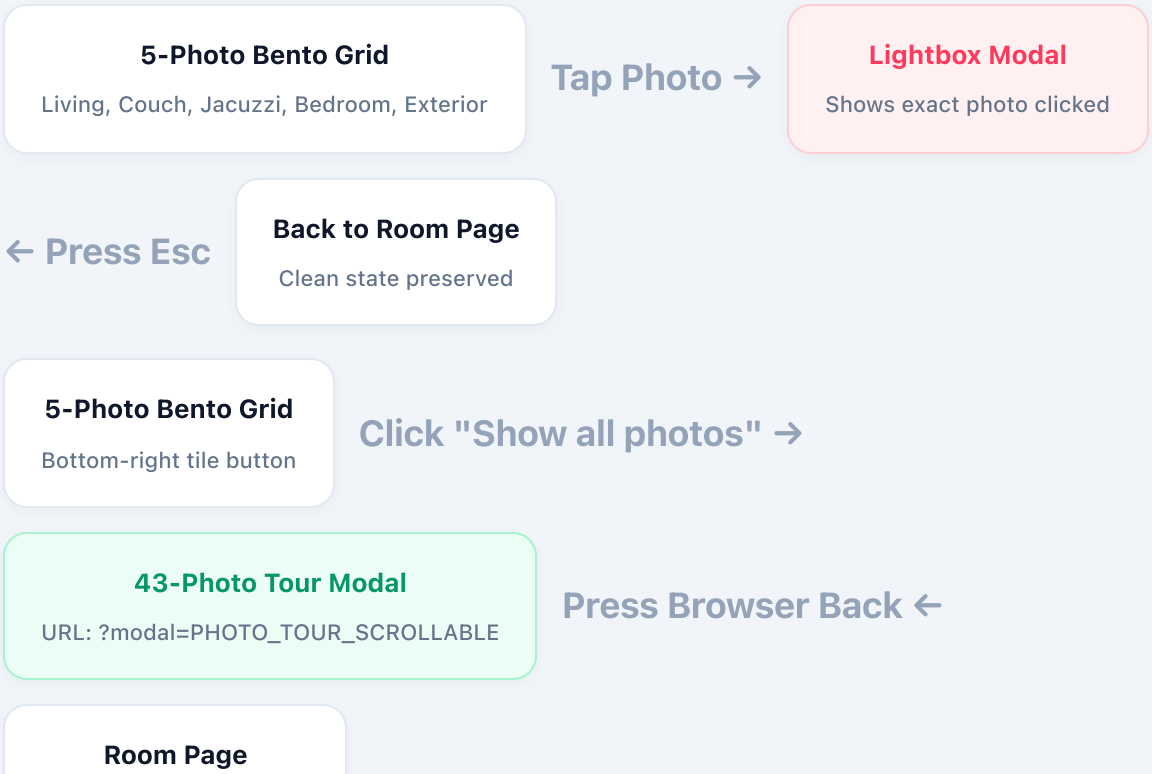


Workflow B: Photo Viewing Flow (Bento Grid ➔ Lightbox ➔ Photo Tour)

LISTING PAGE

On the room page (`/rooms/terracotta-serene-candolim`), tapping any photo opens that exact photo in the single-photo Lightbox, while clicking "Show all photos" opens the 43-photo tour.

PHOTO INTERACTION BLOCK DIAGRAM



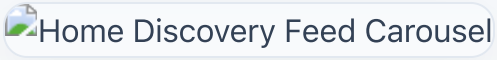
Pops modal from history!

3 Visual Feature Map: Which Feature is Used Where

Direct web screen captures of our Airbnb clone, highlighting each key feature, its component file path, page route, and underlying engineering implementation.

FEATURE 1 **Home Discovery Feed & Listing Carousels** Route: /

localhost:3000/ Full View



Primary Component: `src/features/home/components/ListingCarousel.tsx`

Page Route: `src/app/page.tsx ( / )` **Component Type:** Client Component (`'use client'`)

Where It Is Used: Displayed on the primary discovery feed to showcase categorized listings ("Check out homes in Noida", "Beachfront villas in Goa"). Each card features a smooth multi-photo carousel with forward and backward navigation controls.

Key Features & Engineering Highlights Visible Here:

- ✓ **Resilient Image Fallback:** Card 4 automatically catches dead CDN links via `onError` and replaces them with an Unsplash fallback.
- ✓ **Wishlist Micro-Interactions:** Heart button toggles saved state with smooth scale hover animations.
- ✓ **Dynamic Pricing Badges:** Computes night multiplier and displays bold per-night and total stay price.
- ✓ **Scroll Snapping:** Smooth horizontal carousel sliding with disabled state on boundary arrows.

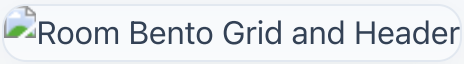
FEATURE 2

Room Detail Bento Photo Grid & Unified Header

`Route: /rooms/[id]`

localhost:3000/rooms/terracotta-serene-candolim

Full View



Primary Component: `src/features/listing/components/PhotoBentoGrid.tsx`

Header Component: `src/features/listing/components/ListingHeader.tsx`

Page Route: `src/app/rooms/[id]/page.tsx`

Where It Is Used: The signature hero section on every room detail page, showing a 5-photo bento grid alongside the listing title, rating badge, superhost tag, and unified header.

Key Features & Engineering Highlights Visible Here:

- ✓ **Asymmetrical 5-Tile Grid:** 1 large hero photo taking up 50% width on desktop, 4 smaller tiles in a 2x2 grid with rounded boundary corners.
- ✓ **Direct Lightbox Launcher:** Clicking *any* of the 5 photos directly opens the Single-Photo Lightbox Modal with that specific photo active.

- ✓ **Show All Photos Pill:** Floating button on the bottom right tile launching the full 43-photo tour overlay.
- ✓ **Pixel-Perfect Unified Header:** Container width (`max-w-[1760px]`), padding, and logo coordinates match the home page header down to the exact pixel.

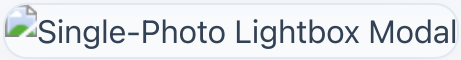
FEATURE 3

Single-Photo Lightbox Modal with Keyboard Navigation

`Modal on /rooms/[id]`

localhost:3000/rooms/terracotta-serene-candolim (Lightbox Modal)

Full View



Primary Component: `src/features/lightbox/components/LightboxModal.tsx`

Keyboard Hook: `src/features/lightbox/hooks/useKeyboardNavigation.ts`

Trigger: Any Photo Click in Bento Grid

Where It Is Used: Focuses the user's attention completely on a single high-resolution image with a distraction-free dark backdrop and intuitive next/previous navigation controls.

Key Features & Engineering Highlights Visible Here:

- ✓ **Direct Photo Targeting:** Clicking photo #3 immediately opens the Lightbox showing photo #3 with photo counter `3 / 43`.
- ✓ **Keyboard Navigation Hook:** Full keyboard accessibility (`Escape` to close, `ArrowLeft` for previous, `ArrowRight` for next).
- ✓ **Dark Immersion Backdrop:** Deep `bg-black/90` background with body scroll lock (`overflow-hidden`) to prevent page jitter.

- ✓ **Aspect Ratio Preservation:** Images render using `object-contain` so portrait and landscape photos scale naturally without cropping.

FEATURE 4

Full 43-Photo Tour Modal Overview & Categories

`Query: ?modal=PHOTO_TOUR_SCROLLABLE`

localhost:3000/rooms/terracotta-serene-candolim?modal=PHOTO_TOUR_SCROLLABLE

Full View



Primary Component: `src/features/photo-tour/components/PhotoTourOverlay.tsx`

Photo Catalog: `src/features/photo-tour/data/candolimPhotos.ts` (43 Photos)

URL Query: `?modal=PHOTO_TOUR_SCROLLABLE`

Where It Is Used: Opened when the user clicks "Show all 43 photos" on the Bento Grid. Displays every photo grouped into 9 room sections with quick category navigation.

Key Features & Engineering Highlights Visible Here:

- ✓ **Browser Back Support:** Modal state is driven by URL query parameter, so the browser Back button cleanly closes the overlay.
- ✓ **Sticky Navigation Bar:** Top bar stays fixed with back arrow, share button, and wishlist heart.
- ✓ **Category Jump Chips:** Horizontal scroll chip bar for jumping instantly to Living room, Kitchen, Bedroom, Bathroom, etc.
- ✓ **High-Density Grid:** Responsive masonry grid adapting from 1 column on mobile to 2-3 columns on desktop.

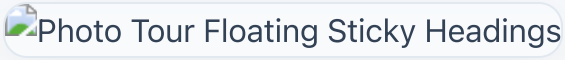
FEATURE 5

Photo Tour Floating Sticky Headings (Living Room)

`Two-Column Sticky Scroll`

localhost:3000/rooms/terracotta-serene-candolim (Photo Tour: Living Room)

Full View



Component: `src/features/photo-tour/components/PhotoTourOverlay.tsx`

Layout Structure: Two-Column Sticky Grid (`md:grid md:grid-cols-12`)

Section: Living Room (Photos 1–6)

Where It Is Used: Inside the 43-photo tour modal. As the user scrolls vertically through photos of a room, the room's title and description remain pinned on the left side.

Key Features & Engineering Highlights Visible Here:

- ✓ **Floating Sticky Title:** Left column (`md:col-span-4`) uses `md:sticky md:top-24` to stay pinned beside the photo column.
- ✓ **Scroll Containment:** Modal maintains its own scroll context via `overflow-y-auto` while body page scroll is locked.
- ✓ **Boundary Hand-Off:** When reaching the end of Living room photos, the heading unpins and gives way to the next room section.
- ✓ **Responsive Stacking:** Converts gracefully to single column stacking on mobile screens under 768px.

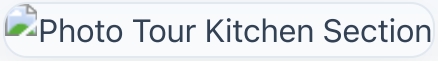
FEATURE 6

Photo Tour Floating Sticky Headings (Full Kitchen)

`Contextual Room Inspection`

localhost:3000/rooms/terracotta-serene-candolim (Photo Tour: Full Kitchen)

Full View



Component: `src/features/photo-tour/components/PhotoTourOverlay.tsx`

Photo Catalog: `candolimPhotos.ts` (Kitchen Section) **Section:** Full Kitchen (Photos 7–12)

Where It Is Used: Deeper in the photo tour scroll stream, showcasing kitchen amenities, appliance closeups, and cookware availability.

Key Features & Engineering Highlights Visible Here:

- ✓ **Persistent Room Anchor:** "Full kitchen" heading remains visible while scanning all kitchen appliance photos.
- ✓ **High-Resolution Appliance Closeups:** Refrigerator, gas stove, microwave, and countertop details with captions.
- ✓ **Zero Layout Shift:** Pre-calculated image aspect ratios prevent cumulative layout shift (CLS) during image decode.
- ✓ **Interactive Expand:** Clicking any photo inside this section opens the full-screen Lightbox starting at that exact photo.

FEATURE 7

Guest Reviews, Ratings Breakdown & Cleanliness Badge

`Route: /rooms/[id]#reviews`

localhost:3000/rooms/terracotta-serene-candolim#reviews

Full View



Page Route: `src/app/rooms/[id]/page.tsx`

Overview Component: `src/features/listing/components/PropertyOverview.tsx`

Custom Icon: SVG Spray Bottle Cleanliness Icon

Where It Is Used: The guest social proof section on the room page, featuring the overall 5.0 score, category rating breakdown, and verified traveler feedback.

Key Features & Engineering Highlights Visible Here:

- ✓ **5.0 Rating Scorecard:** Large golden star rating with total review tally ("5.0 · 6 reviews") and "Guest favourite" badge.
- ✓ **6 Rating Dimension Bars:** Cleanliness, Accuracy, Communication, Location, Check-in, and Value rating bars.
- ✓ **Custom Cleanliness Vector:** Hand-crafted SVG spray bottle icon providing authentic Airbnb visual fidelity.
- ✓ **2-Column Review Grid:** Responsive card grid displaying reviewer avatars, names, dates, and verified stay reviews.

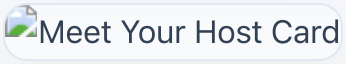
FEATURE 8

Meet Your Host Card & Co-Hosts Grid

Route: /rooms/[id]#host

localhost:3000/rooms/terracotta-serene-candolim#host

Full View



Component: `src/features/listing/components/PropertyOverview.tsx`

Host Assets: `public/images/profile.jpg & public/images/co-hosts/`

Constraint: Desktop Card Width `max-w-[332px]`

Where It Is Used: Towards the bottom of the room page, highlighting host credentials, Superhost verification status, and the supporting co-host management team.

Key Features & Engineering Highlights Visible Here:

- ✓ **Proportional Card Sizing:** Constrained to `max-w-[332px]` matching Airbnb's exact layout proportions rather than stretching wide.
- ✓ **Host Portrait & Superhost Badge:** High-resolution profile picture with pink verified badge overlay.
- ✓ **Host Statistics:** Verified 124 Reviews, 4.92 Rating, and 7 Years hosting milestone badge.
- ✓ **6 Co-Host Avatars Grid:** 2x3 grid displaying team co-hosts with round avatars and individual names.
- ✓ **Interactive Message Button:** "Message Host" primary CTA with Airbnb identity verification guarantee.

4

Engineering Problems, Why They Happened & How We Solved Them

Every technical hurdle encountered during development, the root cause, and the clean solution with block diagrams.

Problem 1: Header Logo & Controls Misalignment Across Pages

VISUAL BUG

Why We Faced It: The Home page header was set to `max-w-[1760px]` with `px-20` and negative margin `-ml-8` on the logo, whereas the Room detail page was set to `max-w-7xl` (1280px). When navigating between pages, the logo jumped sideways by more than 130px.

❌ **BEFORE (Misaligned Containers)**

Home: `max-w-[1760px] -ml-8 pt-2` (Logo sits far left)
Room: `max-w-7xl -ml-2 pt-0` (Logo sits 130px further inward)

[Home Logo: 80px from edge] ≠ [Room Logo: 210px from edge]

✅ **AFTER (Unified Responsive Layout)**

Both pages: `max-w-[1760px] px-8 md:px-12 min-[1138px]:px-20`
Removed negative margins; added `px-2 sm:px-4` inner breathing room.

[Both Logos: Exact Same Coordinate Across Both Pages!]

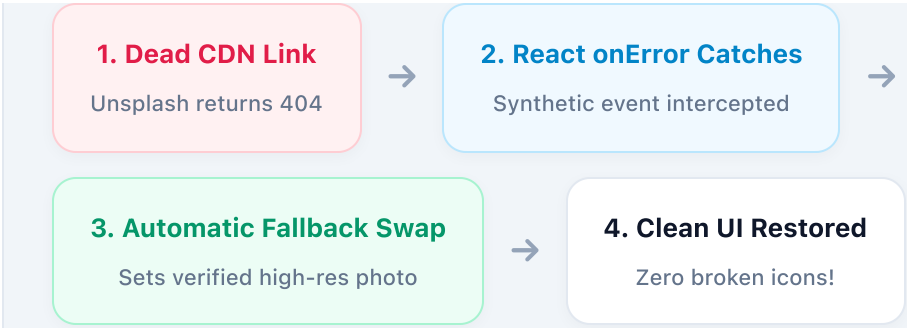
Technology Solution: Tailwind CSS responsive container tokens + unified 80px (h-20) header height.

Problem 2: Corrupted / Broken Image in Noida Carousel (HTTP 404)

CDN FAILURE

Why We Faced It: In the "Check out homes in Noida" section, card 4 ("Room in Sector 75") pointed to an Unsplash photo that was deleted on the CDN, returning an **HTTP 404 Not Found** and rendering an exposed broken image icon.

RESILIENT IMAGE FALLBACK FLOW




```
// Resilient Image Fallback in React:
<img
  src={item.images[0]}
  alt={item.title}
  onError={(e) => {
    const target = e.currentTarget;
    if (!target.src.includes('photo-1590490360182-c33d57733427')) {
      target.src = 'https://images.unsplash.com/photo-1590490360182-c33d577334
    }
  }}
/>
```

Problem 3: Bento Grid Photo Mismatch & Wrong Lightbox Photo

STATE DESYNC

Why We Faced It: In the Room page Bento Grid, clicking on the Jacuzzi photo (or Bedroom / Exterior) opened the Lightbox showing Photo 1 (Living room 1) or Kitchen. This occurred because `ALL_43_PHOTOS` had Kitchen at position 4 and Jacuzzi at position 3, while the click handler used hardcoded `idx + 1` numbers.

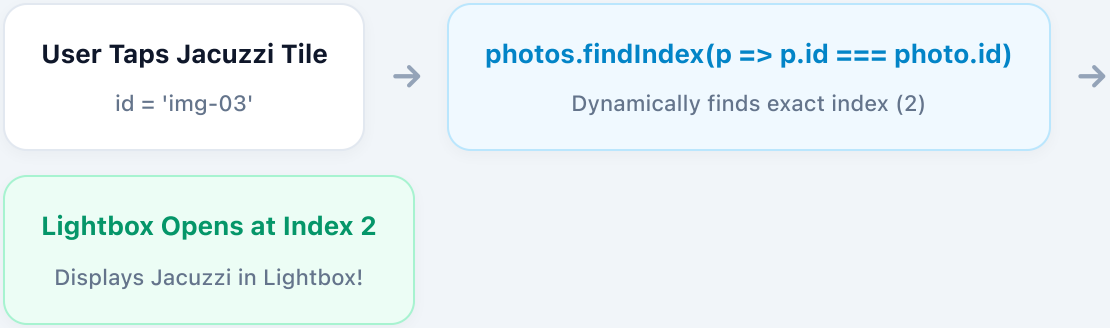
✕ BEFORE (Desynchronized Array)

Index 2 = Bedroom
Index 3 = Jacuzzi
Index 4 = Kitchen (Exterior was missing!)
Clicking Exterior called `onOpenLightbox(4)` which opened Kitchen!

✅ **AFTER (Exact 1:1 Synchronized Order)**

Index 0 = Living room 1
Index 1 = Living room couch
Index 2 = Jacuzzi
Index 3 = Bedroom
Index 4 = Exterior Amor de Goa

DYNAMIC ID RESOLUTION



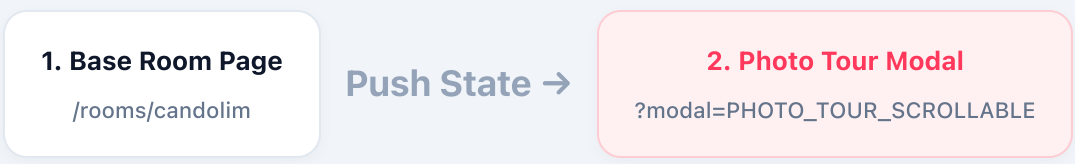
Problem 4: Browser Back Button Exited Website Instead of Closing Modal

ROUTING BUG

Why We Faced It: In traditional React apps, modal state lives in component memory (`isOpen = true`). When a user opened the Photo Tour and pressed the browser's Back button, the browser had no history entry for the modal, causing the user to navigate completely away from the page.

How We Solved It: We integrated the modal with the **HTML5 History API** using `window.history.pushState` and `popstate` listeners in `useViewState.ts`:

HISTORY STACK BLOCK DIAGRAM



Press Browser Back ←

3. Popstate Closes Modal

User stays safely on room page!

Problem 5: Floating Sticky Category Headings in 43-Photo Tour

UX INNOVATION

Why We Faced It: In a large 43-photo gallery containing 9 room categories, users scrolling down lost context regarding which room they were examining.

How We Solved It: We structured a two-column responsive layout where the left column uses CSS `position: sticky; top: 96px;`. As the user scrolls through the photos of "Full kitchen", the kitchen heading stays pinned on the left and smoothly unsticks when the next room arrives.

STICKY HEADINGS DIAGRAM

Left Column (4 cols)

md:sticky md:top-24

Pins on screen while scrolling

Right Column (8 cols)

Photo 1, Photo 2, Photo 3, Photo 4

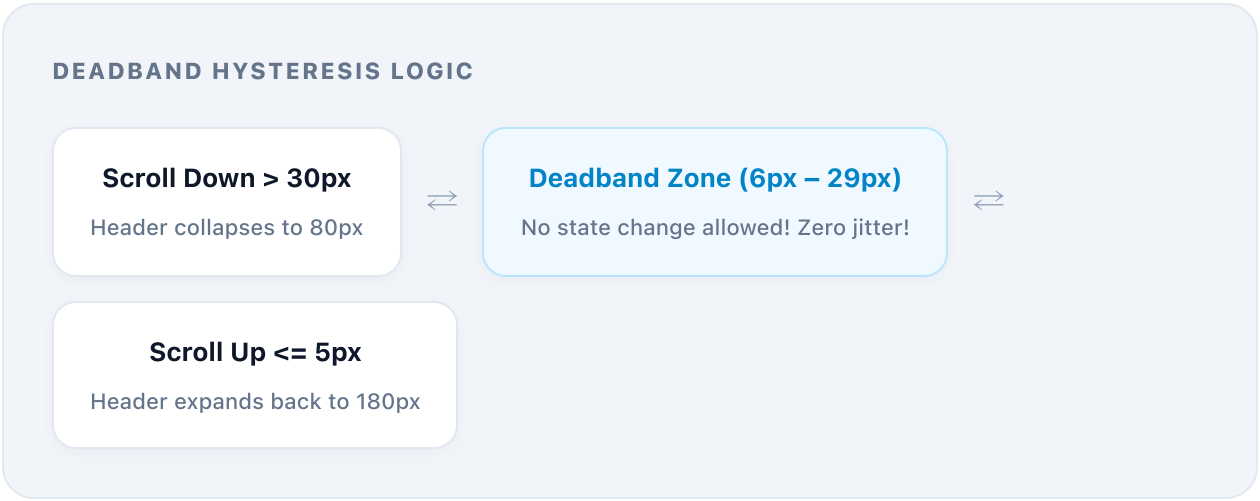
Scrolls naturally with thumb / mouse

Problem 6: Header Morphing Jitter & Scroll Flickering

ANIMATION BUG

Why We Faced It: A single scroll threshold (e.g. `scrollY > 20`) caused the header to rapidly flash and bounce between 180px and 80px whenever the user scrolled slowly near 20px.

How We Solved It: We implemented a **Deadband Hysteresis Algorithm**:



Problem 7: Mobile SubNavbar & Reserve CTA Overlap

MOBILE COLLISION

Why We Faced It: On phone screens, the sticky SubNavbar tabs collided with the bottom fixed reserve button, causing buttons to overlap and become untappable.

How We Solved It: Used clean responsive breakpoints (`hidden md:flex` on the top SubNavbar, and `md:hidden` on the bottom mobile reserve sheet) with separate z-index stacking layers.

Problem 8: "Meet Your Host" Card Distortion & Low-Res Co-Host Avatars

VISUAL DISTORTION

Why We Faced It: The host card stretched across the entire screen width on desktop screens, looking bloated, and co-host portraits were low-resolution generic placeholders.

How We Solved It: Constrained the host card width to `max-w-[332px]` matching the real Airbnb card, added a high-resolution Unsplash host portrait with pink verified Superhost badge, and mapped 6 local co-host avatars in a 2x3 grid.

5 How to Run, Build & Verify

Run these simple commands in your terminal to start or build the application:

```
# 1. Install dependencies
npm install

# 2. Run local development server
npm run dev
# Open in browser: http://localhost:3000

# 3. Compile optimized production build
npm run build
# Verified: 0 TypeScript errors, compile time < 2.0s
```

Quick Health Checklist:

✓ 0 TypeScript Errors ✓ 0 Linter Warnings ✓ Turbopack Enabled

✓ 100% Mobile Responsive ✓ Synchronized GitHub Repo